

MAHADEVU M P

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Summary

Detail-oriented Software Engineering graduate with practical experience in manual testing concepts, STLC, defect life cycle management, and API testing using Postman through automation and data engineering projects. Demonstrated ability to create and execute test cases, perform bug reporting and defect tracking, and ensure product quality throughout the SDLC.

Skills

Programming & Querying: SQL, Python, C

Modern Data Stack: AWS, Snowflake, dbt, ETL Pipelines, Data Modeling

Data Visualization Tools: MS Excel, Power BI

Tools: Git, GitHub, UIPath, JN, MySQL, MS Office

Projects

IoT-Based Automatic Water Monitoring System (P) | *IoT, Blynk, Python, ML* **Nov 2024 – Jul 2025**

- Designed and developed a real-time IoT-based water monitoring system using ESP32, ultrasonic/level sensors, and Blynk for continuous upper and lower tank level tracking with automated motor control.
- Integrated machine learning models (DBSCAN, LSTM) for water consumption pattern analysis, anomaly detection, and predictive insights to optimize usage and prevent overflow/dry-run conditions.
- Ensured system reliability through threshold-based alerts, notification mechanisms, and performance optimization for low-power and real-time operations.

Airbnb Data Engineering Pipeline | *Snowflake, dbt, AWS, Python* **Jan 2026 – Feb 2026**

- Built a scalable Medallion Architecture pipeline (Bronze → Silver → Gold) to process Airbnb listings, bookings, and hosts data from AWS S3 into Snowflake, ensuring structured and analytics-ready datasets.
- Implemented dbt transformations including incremental models and SCD Type 2 snapshots, while designing fact tables and One-Big-Table (OBT) models with lineage tracking, testing, and dependency management.
- Optimized data pipelines using Snowflake warehousing, clustering, and ephemeral models while enforcing data quality checks (unique, not-null, referential integrity, business rules) for efficient and trustworthy analytics.

Advanced Sales & Shipment Analytics Dashboard | *Power BI, DAX, Power Query, Data Modeling*

- Designed and developed an interactive Power BI dashboard to analyze global sales, shipments, cost, and profitability for a chocolate manufacturer (Awesome Chocolates), tracking KPIs such as total sales, profit %, boxes shipped, and shipment count across regions, products, and salespeople.
- Built a multi-table data model (shipments, calendar, products, people, locations) and wrote 15+ DAX measures using DIVIDE, PREVIOUSMONTH, PREVIOUSYEAR, and LASTDATE to deliver MoM/YoY growth tracking, latest-month summaries, and profit margin analysis.
- Enhanced usability with a dynamic measure selector, region slicers, bookmark navigation, custom tooltips, a salesperson leaderboard, and a gauge for light-box (under 50 boxes) shipments, helping business users identify top-performing regions and reps and monitor cost efficiency.

Docent: Context-Aware AI Document Assistant | *Next.js, FastAPI, LangChain, ChromaDB, Docker*

- Designed and developed a full-stack RAG document assistant in 7 days using AI-assisted development, letting users upload PDF/Markdown files and get answers grounded only in that document, with clickable inline citations that jump to the exact source page or heading.
- Built the retrieval pipeline (text extraction → LangChain chunking → embeddings → ChromaDB) with page/heading metadata on every chunk, streaming answers over SSE, and a fallback that skips the LLM when nothing relevant is retrieved to prevent hallucination, with swappable OpenAI, Anthropic, and Gemini providers.
- Implemented a secure FastAPI backend (10+ REST endpoints) with bcrypt/JWT and Google OAuth authentication, per-user data isolation, prompt-injection-resistant prompts, and sanitized uploads, then containerized the frontend, backend, and ChromaDB as 3 Docker Compose services and deployed the demo on Render.

Certifications

NPTEL Certifications 2025 – Python Programming, IoT & Cloud Computing

[View credentials](#)

Cyber Security Fundamentals – Zscaler

[View credentials](#)

Published Paper – IoT-Based Automatic Water Monitoring System

[View credentials](#)

Education

B.E. Information Science & Engineering

Graduated: 2025

P.E.S. College of Engineering, Mandya

CGPA: 7.8 / 10